

Kamal Joshi

Senior Software Engineer II — Backend Platform & Distributed Systems | Go | AWS | Kubernetes | Event-Driven Architecture | Observability

Austin, TX | Open to remote-friendly and hybrid roles

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PROFESSIONAL SUMMARY

Senior backend engineer and technical leader with deep experience building, scaling, and operating highly available distributed systems and cloud-native platform services — from architecture and API design through production incident recovery and multi-region rollouts. Consistent track record of driving projects end to end, making architecture decisions that communicate tradeoffs clearly, and raising the bar for engineering teams through mentorship, observability improvements, and platform tooling. Effective at directing AI tools (Claude Code, GitHub Copilot, Codex) across the full SDLC to move fast while owning the design and quality bar.

CORE STRENGTHS

- **Distributed systems & event-driven architecture:** Go, Java, REST, gRPC, microservices, event-driven pipelines, SQS/SNS, DynamoDB Streams, CloudEvents, high-throughput message delivery, fault tolerance, consistency tradeoffs
- **Cloud & containers:** AWS (Lambda, ECS, EKS, S3, SQS/SNS, DynamoDB, Secrets Manager), Kubernetes, Docker, multi-region deployments, sovereign/GovCloud environments
- **Infrastructure as Code & CI/CD:** Terraform modules, Terraform pipelines, multi-account infrastructure, deployment validation, automated rollouts
- **Observability & operational excellence:** metrics instrumentation, alerting, DLQ monitoring, incident response, postmortems, retention-risk alerting, rate-limit governance
- **Security & access:** Kong API Gateway, Authorization Service, OAuth 2.0, OIDC, OPA policy enforcement, API key management, token lifecycle
- **Technical leadership:** architecture design, project ownership, cross-team coordination, mentoring, code and design reviews, pragmatic tradeoff decisions
- **AI-assisted SDLC:** Claude Code, GitHub Copilot, Codex — daily use across implementation, IaC, testing, debugging, and code review; pioneered team-wide AI adoption

EXPERIENCE

Cisco

Software Engineering Technical Leader

November 2025 - Present | Austin, TX

- Owned and led design, implementation, and production operation of **API Gateway, Authorization Service, and Third-Party Integrations Service** — cloud-native, microservices-based backend systems on AWS serving enterprise and government customers across multiple regions.
- Drove architecture decisions balancing reliability, scalability, and operability for a real-time **event-driven delivery pipeline** (webhooks, SQS/SNS, DynamoDB Streams, CloudEvents) — communicating tradeoffs across product and engineering stakeholders.
- Prevented service disruption during a production incident by identifying a **300k+ message DLQ backlog**, quantifying blast radius across **15 orgs**, recovering within **5-10 minutes**, and driving systemic observability and retention-risk alerting improvements.
- Drove multi-region production readiness by decomposing a high-risk **Terraform** rollout into staged changes across ECR, Secrets Manager, DynamoDB Streams, SNS, service config, and audit logging — validating **HA behavior across multiple regions**.
- Resolved a customer-scale onboarding failure by tracing **250+ appliance registration requests** through API key authentication and APIGW rate-limit behavior, unblocking the customer and driving durable fixes around retry/backoff, token reuse, and service ownership.

- Led design and delivery of real-time security event push using the open **CloudEvents** standard, delivering **alpha on schedule (Nov 2025)** with a team of three engineers.
- Reframed API onboarding as a control-plane problem and proposed a hybrid Kong Admin API + Terraform model — a platform productivity improvement that improved self-service speed while preserving auditability.

Senior Software Engineer

January 2021 - November 2025 | Austin, TX

- Implemented **metrics instrumentation** for the Authorization Service and Integrations Service; owned production observability, alerting, and DLQ monitoring across identity-adjacent backend services.
- Built **auth-tools**, an internal library adopted across multiple platform services to simplify OAuth 2.0 / policy enforcement integration — a reusable platform component that reduced integration friction team-wide.
- Mentored early-in-career engineers through code review and design feedback; initiated **“Tip Talk Tue”**, a recurring team forum on engineering productivity and best practices.
- Integrated AI tooling (Claude Code, Copilot, Codex) into the full SDLC; **pioneered team-wide GitHub Copilot adoption** ahead of broader Cisco initiatives.

Randstad Technologies US at Cisco

Senior Software Developer

March 2015 - January 2021 | Dublin, OH

- Designed and implemented **Go-based gRPC and REST APIs** for a cloud-native data ingestion platform; used OpenTSDB as the timeseries database for platform metrics.
- Built API-driven **OIDC and OAuth 2.0** authentication and authorization flows for distributed services.
- Contributed to automatic generation of protocol buffer specs and gRPC APIs from SQL schemas, improving API consistency across services.
- Reduced **100+ OWASP-reported security warnings to zero** through secure coding, browser security headers, input sanitization, and server-side validation.

Earlier Engineering Experience

Cognizant, Oracle, Collabera | December 2007 - February 2015

- Delivered backend services, REST APIs, and integration features across financial services, healthcare, insurance, and public-sector domains.
- Built a foundation in Java, REST APIs, SQL, distributed service patterns, and maintainable engineering practices.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate
- AWS Certified Developer - Associate
- MongoDB Certified Java Developer M101J
- Google Cloud Platform Essentials

EDUCATION

Bachelor of Engineering, Information Technology
Dharmsinh Desai University